

Module-I: Wave Optics

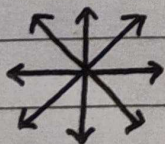
Polarization

In natural light, the optical vector (electric vector) has no specific orientation. It can vibrate in any direction in a plane perpendicular to the direction of propagation of light wave.

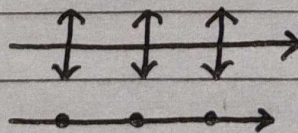
When it is passed through certain crystals (i.e. Quartz, Tourmaline, Calcite, etc), the light coming out of the crystal has optical vectors only in one specific direction. This light is called polarized light.

"The phenomena of changing unpolarized (or natural) light into polarized light is called polarization."

A plane containing vibrations is called plane of vibration while the other plane perpendicular to it which has no vibration is called plane of polarization.



Unpolarized Light



Polarized Light

Polarization By Double Refraction

The phenomena of causing two refracted rays is called double refraction or birefringence.

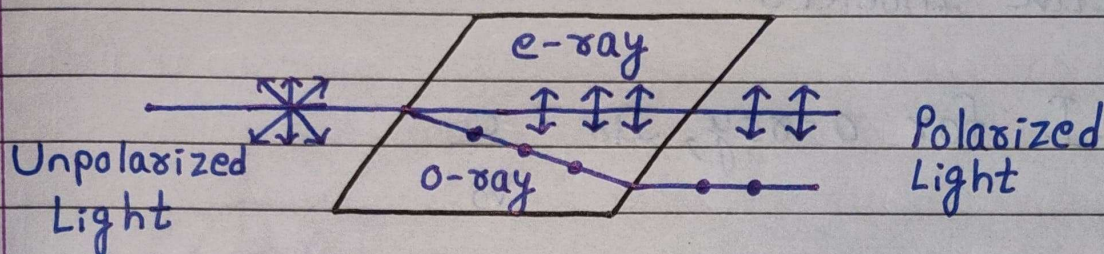


Fig.: Calcite (CaCO_3) Crystal
[Birefringent Crystal]

When unpolarized light incident on calcite crystal, it splits into two rays. One of the ray obeys Snell's law, this ray is called ordinary ray or o-ray and the other ray does not obey Snell's law, this ray is called extra-ordinary ray or e-ray.

Both the o-ray & e-ray are linearly (or plane) polarized but their polarization is perpendicular to each other.

Properties of o-ray & e-ray:

1. o-ray obeys Snell's law while e-ray does not obey.
2. Both rays are polarized but are in mutually perpendicular directions.

3. Velocity of o-ray is same in all directions within the crystal while velocity of e-ray is different in different directions.

4. Refractive Indices:

$$\text{R.I. for o-ray, } \mu_o = \frac{c}{v_o}$$

$$\text{R.I. for e-ray in the crystal,} \\ \mu_e = \frac{c}{(v_e)_{\min}}$$

$$\text{R.I. for e-ray in -ve crystal,} \\ \mu_e = \frac{c}{(v_e)_{\max}}$$

5. Let 'd' be the thickness of crystal.

$$\text{Optical path for o-ray} = \mu_o d$$

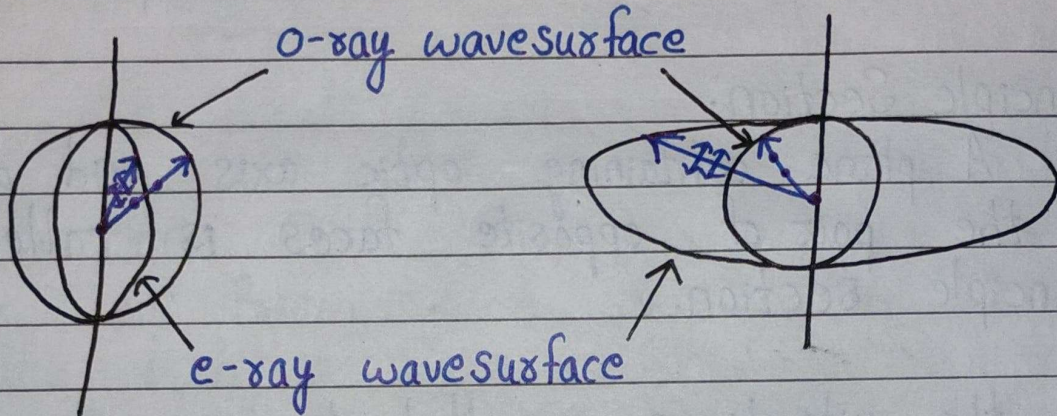
$$\text{Optical path for e-ray} = \mu_e d$$

$$\text{Path difference} = (\mu_e - \mu_o) d$$

$$\text{Phase difference} = \frac{2\pi}{\lambda} (\mu_e - \mu_o) d$$

6. The wavefront (or wave-surface) of o-ray is spherical and wavefront of e-ray is elliptical.

For +ve crystal, ellipse is inside the sphere while for -ve crystal, sphere is inside the ellipse.



+ve crystal [$\mu_e > \mu_o$]

-ve crystal [$\mu_e < \mu_o$]

7. The distinction of o-ray & e-ray exists only within the crystal. Outside the crystal both rays behave as polarized light.

Geometry of Calcite Crystal

The crystal has rhombohedral shape bounded by six faces. Each face is a parallelogram with angles 78° & 102° .

(i) Blunt Corners:

Two corners where ~~are~~ all three angles are obtuse (102°), are called blunt corners. These two blunt corners are diagonally opposite to each other.

mp: (ii) Optic Axis:
Any line bisecting the blunt corner gives the direction of optic axis. A ray incident in this direction do not split into e-ray & o-ray.

"Optic Axis is a direction not a particular line."

(iii) Principle Section:

A plane containing optic axis and perpendicular to the pair of opposite faces is called principle section.

All the vibrations parallel to principle section make e-ray and all the vibrations perpendicular to principle section make o-ray.

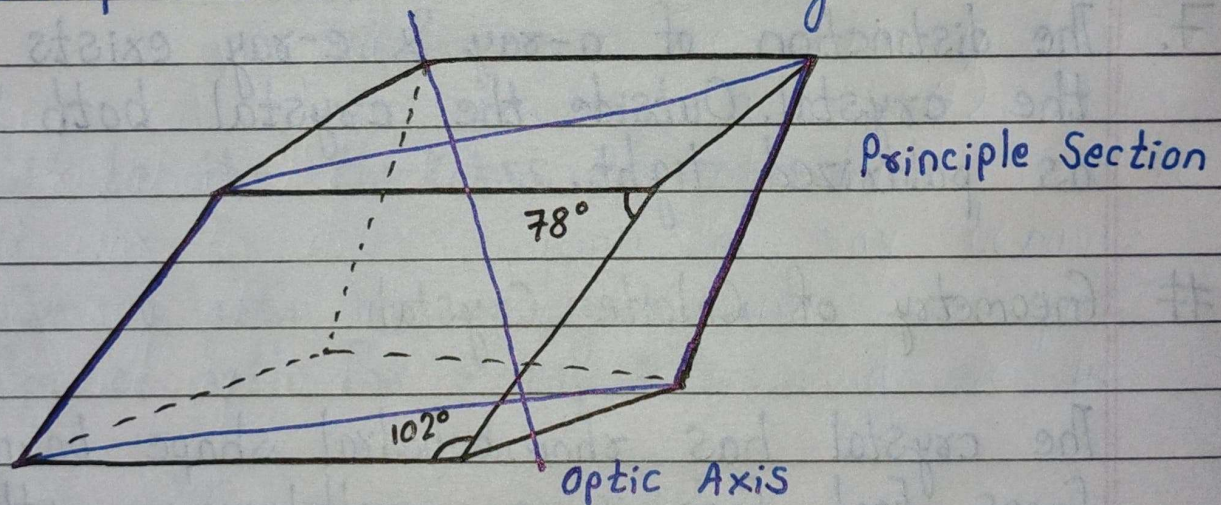


Fig.: Calcite Crystal

Nicol Prism

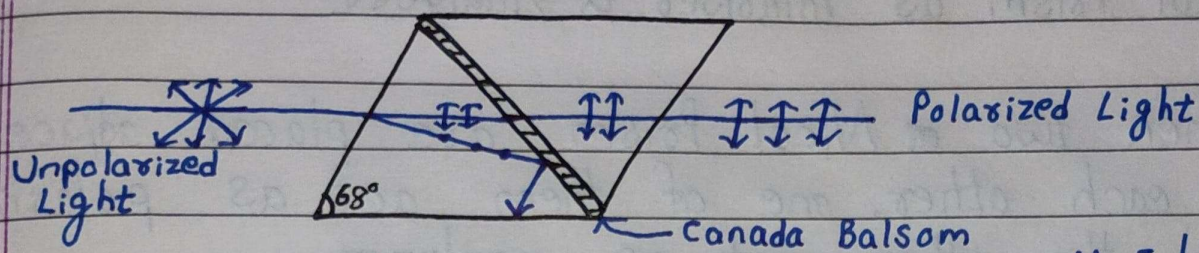


Fig.: Calcite Crystal

$$\mu_e = 1.486$$

$$\mu_c = 1.55$$

$$\mu_o = 1.66$$

Construction:

Nicol Prism is a rhombohedral shaped calcite crystal which is three times long as it is thick. The end surfaces of crystal are grounded and polished until they make an angle of 68° .

This piece is cut into two parts along a plane perpendicular to principle section and new end surfaces. This two points are joint together with Canada Balsom, whose refractive index (μ_c) lies between refractive index of crystal for e-ray and o-ray.

Working:

When unpolarized light enters the calcite, it split into e-ray & o-ray. The angle of incidence on canada balsom and its refractive indices are such that the o-ray suffers total internal reflection while e-ray crosses the canada balsom layer.

The surface where o-ray incidence is blackened to absorb all the o-rays. Thus only e-ray

comes out of crystal, which is linearly polarized.

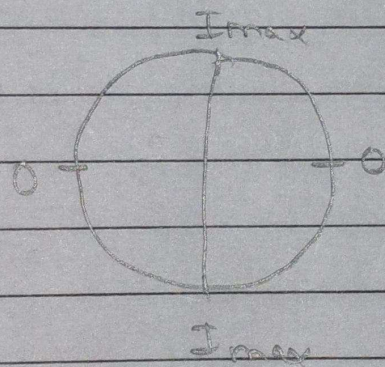
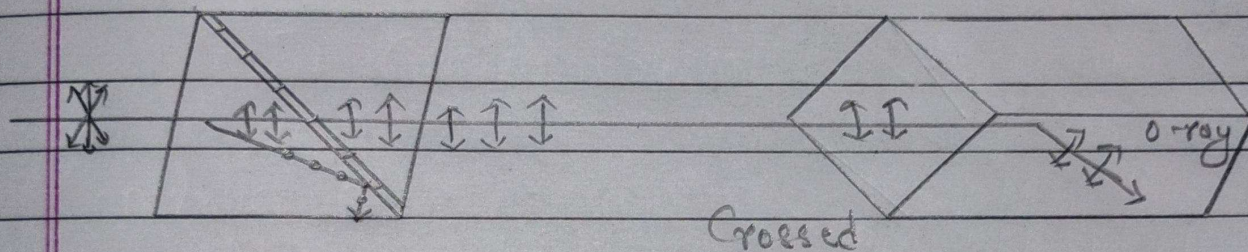
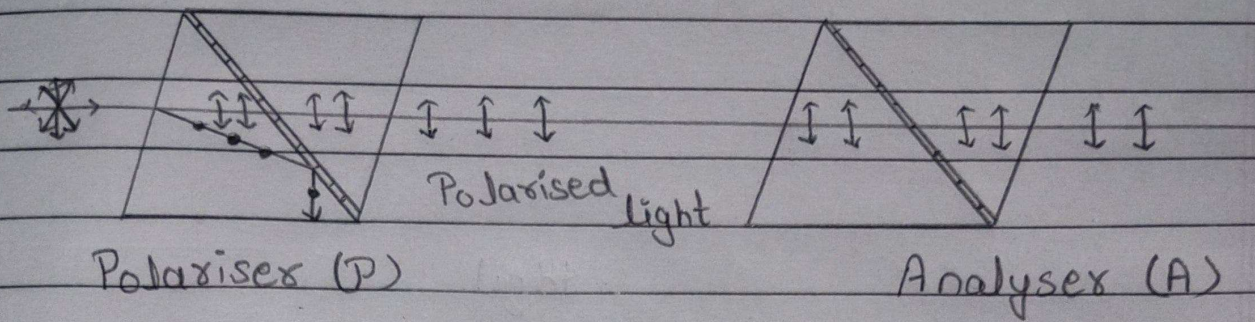
Nicol Prism as Polariser & Analyser

When two Nicol Prism are placed adjacent to each other, one of them act as polarizer and the other act as analyzer.

When unpolarized light incident on polarizer, only e-ray comes out which is polarized. This ray enters prism A whose principle section is parallel to that of P. Hence it is transmitted through analyzer and we get maximum intensity.

Now, A is rotated slowly. The intensity decreases according to Malus Law ($I = a^2 \cos^2 \theta$). After 90° rotation, the vibration of light will be perpendicular. In this position, the ray behaves as o-ray. It suffers TIR and we get zero intensity. Both prism are said to be crossed.

On rotating the analyzer through 360° , two points of maximum intensity and two points of zero intensity are obtained for linearly polarized light.



Theory of Plane, Elliptically & Circularly Polarized Light

Consider two linearly polarized light waves having vibration on X-axis & Y-axis.

$$x = a \sin(\omega t + \delta) \quad \text{--- (1)}$$

$$y = b \sin \omega t \quad \text{--- (2)}$$

$$\Rightarrow \sin \omega t = \frac{y}{b}$$

$$\cos \omega t = \sqrt{1 - \frac{y^2}{b^2}}$$

From eqⁿ. (1),

$$\frac{x}{a} = \sin \omega t \cdot \cos \delta + \cos \omega t \cdot \sin \delta$$

$$\Rightarrow \frac{x}{a} = \frac{y}{b} \cdot \cos \delta + \sin \delta \cdot \sqrt{1 - \frac{y^2}{b^2}}$$

$$\Rightarrow \frac{x}{a} - \frac{y}{b} \cos \delta = \sin \delta \sqrt{1 - \frac{y^2}{b^2}}$$

Squaring both sides, we get,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \cos^2 \delta - \frac{2xy}{ab} \cos \delta = \sin^2 \delta - \frac{y^2}{b^2} \sin^2 \delta$$

$$\Rightarrow \boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} \cos \delta = \sin^2 \delta}$$

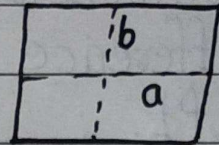
Case-I: $\delta = 0^\circ$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab} = 0$$

$$\Rightarrow \left(\frac{x}{a} - \frac{y}{b} \right)^2 = 0$$

$$\Rightarrow \frac{x}{a} = \frac{y}{b}$$

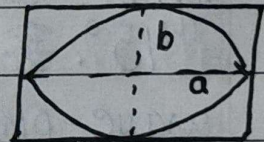
$\Rightarrow y = \frac{b}{a} \cdot x$, Eqⁿ of straight line.



Plane Polarized Light (PPL):

A polarized light having vibrations along a straight line is called PPL.

Case-II: $\delta = \pi/2$



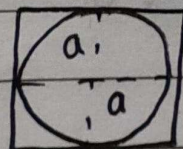
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \text{ Eqⁿ of ellipse.}$$

Elliptically Polarized Light (EPL):

Combination of two linear vibration having 90° phase difference and different amplitude produce EPL.

Case-III: $\delta = \pi/2$ & $a = b$

$$\frac{x^2}{a^2} + \frac{y^2}{a^2} = 1$$



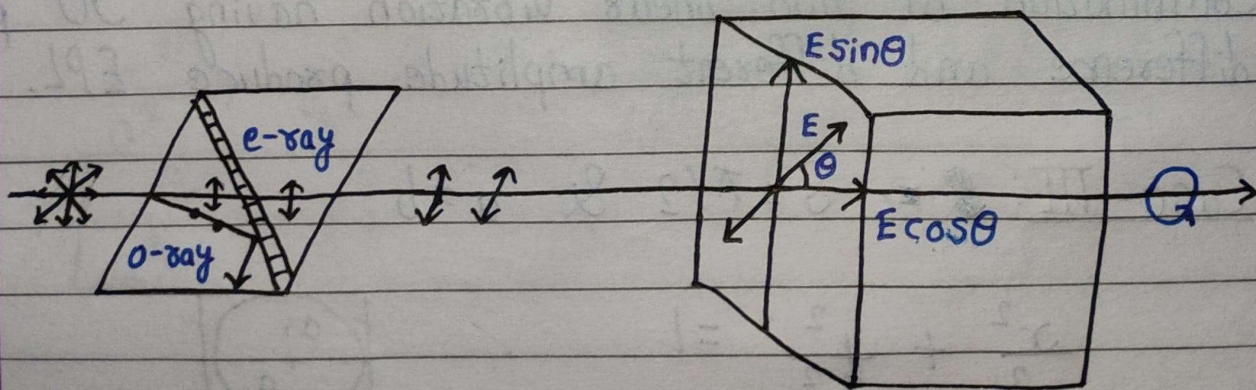
$$\Rightarrow x^2 + y^2 = a^2, \text{ Eqn. of circle.}$$

Circularly Polarized Light (CPL):

Combination of two linear vibrations, having phase difference 90° and equal amplitude produces CPL.

Production of PPL, EPL & CPL:

1. In Nicol Prism, eliminated due TIR. Only e-ray comes out which is plane polarized.
2. Polarized light is allowed to incident on quarter wave-plate. The angle θ between vibration of light and optic axis of quarter wave-plate is 45° . ~~The~~ Two components will be equal. Quarter wave-plate introduced a phase difference of 90° between them. They combine and form CPL.
3. In this case, θ is not equal to 0° , 45° or 90° . So two components will be different. They will combine and form EPL.



Quarter Wave Plate

Detection of PPL, EPL & CPL:

The light under test is passed through Nicol Prism, and prism is rotated by 360° .

1. If intensity changes from maximum to zero, then light is PPL.
2. If intensity changes from maximum to minimum but never zero, then light is EPL.
3. If no change in intensity observed, then light is CPL.

Optical Activity (Rotatory Polarization)

When polarized light is passed through certain substances, their plane of polarization is rotated by angle θ , this is called optical activity.

Optically active substances are of two types:

1. Dextro-Rotatory (or Right Handed):
It rotates the vibration clock-wise. For example, sugar, camphor, etc.
2. Laevo-Rotatory (or Left Handed):
It rotates the vibration anti-clockwise. For example, fructose, cholesterol, etc.

According to Biot's Law:

- (i) $\theta \propto l$
- (ii) $\theta \propto C$ (Concentration)
- (iii) $\theta \propto \frac{1}{\lambda^2}$
- (iv) In a mixture,
 $\theta = \theta_1 + \theta_2 + \theta_3 + \dots$ $\begin{array}{l} \text{anti-clockwise} \rightarrow +ve \\ \text{clockwise} \rightarrow -ve \end{array}$

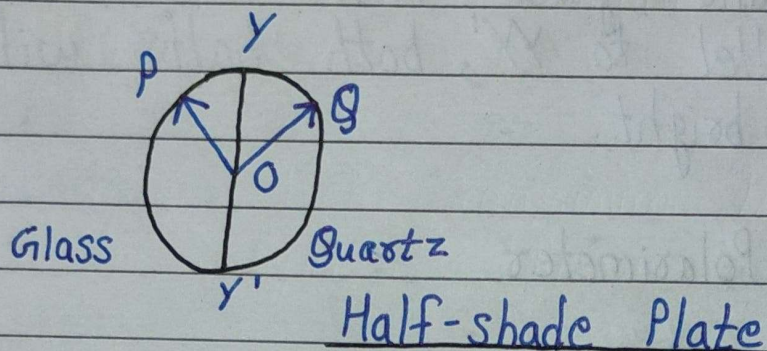
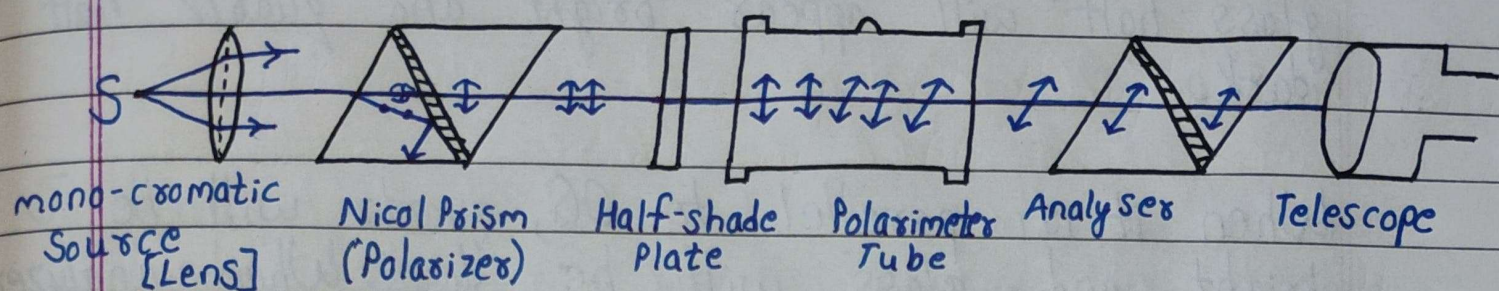
Specific Rotation:

It is defined as amount of rotation produced by 10 cm long substance, having concentration 1 gm/cc.

$$\boxed{[S]_{\lambda}^T = 10 \cdot \frac{\theta}{lc}} \quad \left[\begin{array}{l} \theta \propto lc \\ \theta = slc \end{array} \right]$$

$$\boxed{[l \rightarrow \text{cm} \ \& \ C \rightarrow \text{gm/cc}]}$$

Laurent's Half Shade Polarimeter

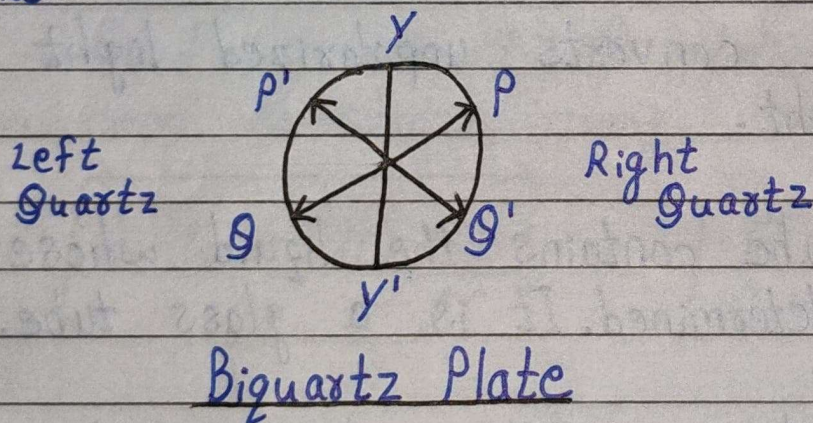
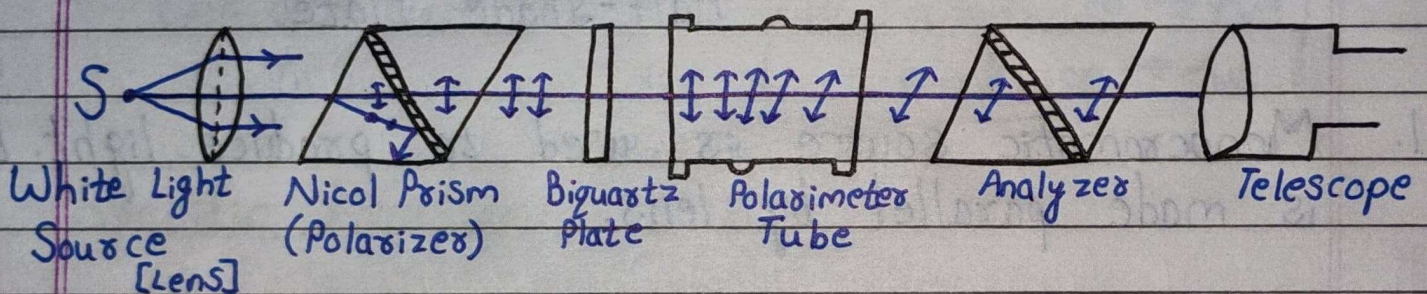


1. Monochromatic source is used to produce light. Light is made parallel by lens.
2. Nicol prism converts unpolarized light into polarized light.
3. Polarimeter tube contains the liquid whose rotation is to be determined. It is a glass tube.
4. Another Nicol prism is used as analyser to detect the plane of vibration of light. Telescope is used to see the light.
5. Half-shade plate is a circular plate. Half part is made up of glass and half part is made by quartz. If the incident light has vibration along OP , glass half will not affect the direction of vibration. The light passing from quartz will have vibration along OQ because it is dextro-rotatory.

When analyzer of principle section is along OP , glass half will appear bright and quartz half dark.

When it is parallel to OQ , quartz will be bright and glass will be dark. When analyzer is parallel to YY' , both halves will appear equally bright.

Biquartz Polaximeter



1. White light source is used to produce light. Light is made parallel by lens.
2. Nicol Prism converts unpolarized light into polarized light.
3. Polaximeter tube contains the liquid whose rotation is to be determined. It is a glass tube.

4. Another Nicol Prism is used as analyzer to detect the plane of vibration of light. Telescope is used to see the light.
5. Biquartz plate is made-up of left handed & right handed quartz. If direction of incident light vibration is along POQ , then in right handed plate it is rotated to OQ' and in left handed plate, it is rotated to OP' .

Therefore, one half appears red and other half appears blue. When principle of analyzer is parallel to YY' , both half appears grey. This position is called Tint of Passage.

Numericals

Q) A tube 20 cm long with a solution of 15 gm cane sugar dissolved in 100 cc of water is placed in the path of polarized light. Find the angle of rotation, if the specific rotation of cane-sugar is $66^\circ (\text{cm}^{-1})(\text{gm/cc})^{-1}$.

Solⁿ:

$$l = 20 \text{ cm}$$

$$m = 15 \text{ gm}$$

$$V = 100 \text{ cc}$$

$$S = 66^\circ (\text{cm})^{-1} (\text{gm/cc})^{-1}$$

Now,

$$c = \frac{m}{V} = \frac{15}{100} = 0.15 \text{ gm/cc}$$

We have,

$$S = 10 \cdot \frac{\theta}{lc}$$

$$\Rightarrow \theta = \frac{S lc}{10} = 19.8^\circ.$$

Q) A sugar solution in a tube of 20 cm produces optical rotation of 13° . The solution is then diluted to $1/3$ of its previous concentration. Find the optical rotation produced by 30 cm long tube containing the diluted solution?

Solⁿ:

$$l_1 = 20 \text{ cm}$$

$$l_2 = 30 \text{ cm}$$

$$\theta_1 = 13^\circ$$

$$C_1 = C$$

$$C_2 = C/3$$

$$\theta_2 = ?$$

We have,

$$S = \frac{l_1 \cdot \theta_1}{l_1 \cdot C_1} = \frac{l_2 \cdot \theta_2}{l_2 \cdot C_2}$$

$$\Rightarrow \theta_2 = \frac{\theta_1 \cdot l_2 \times 1/3}{l_1} = \frac{30 \times 1/3 \times 13}{20} = 6.5^\circ$$

Q) A 5% solution of cane-sugar placed in a tube of length 40 cm causes the optical rotation of 20° . How much length of 10% solution of the same substance will cause 35° rotation.

Solⁿ:

$$l_1 = 40 \text{ cm}$$

$$l_2 = ?$$

$$\theta_1 = 20^\circ$$

$$\theta_2 = 35^\circ$$

$$C_1 = 5\% \text{ of } c = 0.05 c$$

$$C_2 = 10\% \text{ of } c = 0.1 c$$

We have,

$$S = \frac{l_1 \cdot \theta_1}{l_1 C_1} = \frac{l_2 \cdot \theta_2}{l_2 C_2}$$

$$\Rightarrow l_2 = \frac{\theta_2 \times l_1 \times 0.05}{0.1 \times \theta_1} = 35 \text{ cm.}$$

Q) 80 gm of impure sugar is dissolved in one litre of water. The solution gives an optical rotation of 9.9° when placed in a tube of length 20 cm. If the specific rotation of pure sugar is $66^\circ (\text{cm})^{-1} (\text{gm/cc})^{-1}$, Find the percentage of purity of sugar sample?

Solⁿ:

$$\theta = 9.9^\circ$$

$$l = 20 \text{ cm}$$

$$S = 66^\circ (\text{cm})^{-1} (\text{gm/cc})^{-1}$$

$$C_{\text{impure}} = \frac{80}{1000} \text{ gm/cc} = 0.08 \text{ gm/cc}$$

$$S = \frac{10 \cdot \theta}{l C_{\text{pure}}}$$

$$\Rightarrow C_{\text{pure}} = \frac{10 \times \theta}{S l} = \frac{10 \times 9.9}{66 \times 20} = 0.075 \text{ gm/cc}$$

$$\% \text{ purity} = \frac{C_{\text{pure}}}{C_{\text{impure}}} \times 100 = 93.75 \%$$